

Dietary fiber intake and risk of breast cancer: a meta-analysis of prospective cohort studies.

BACKGROUND: Observational and preclinical studies suggest that dietary fiber intake may reduce the risk of breast cancer, but the results are inconclusive. **OBJECTIVE:** We aimed to examine the association between dietary fiber intake and risk of breast cancer by conducting a meta-analysis of prospective cohort studies. **DESIGN:** Relevant studies were identified by a PubMed database search through January 2011. Reference lists from retrieved articles were also reviewed. We included prospective cohort studies that reported RRs with 95% CIs for the association between dietary fiber intake and breast cancer risk. Both fixed- and random-effects models were used to calculate the summary risk estimates. **RESULTS:** We identified 10 prospective cohort studies of dietary fiber intake and risk of breast cancer involving 16,848 cases and 712,195 participants. The combined RR of breast cancer for the highest compared with the lowest dietary fiber intake was 0.89 (95% CI: 0.83, 0.96), and little evidence of heterogeneity was observed. The association between dietary fiber intake and risk of breast cancer did not significantly differ by geographic region, length of follow-up, or menopausal status of the participants. Omission of any single study had little effect on the combined risk estimate. Dose-response analysis showed that every 10-g/d increment in dietary fiber intake was associated with a significant 7% reduction in breast cancer risk. Little evidence of publication bias was found. **CONCLUSION:** This meta-analysis provides evidence of a significant inverse dose-response association between dietary fiber intake and breast cancer risk.